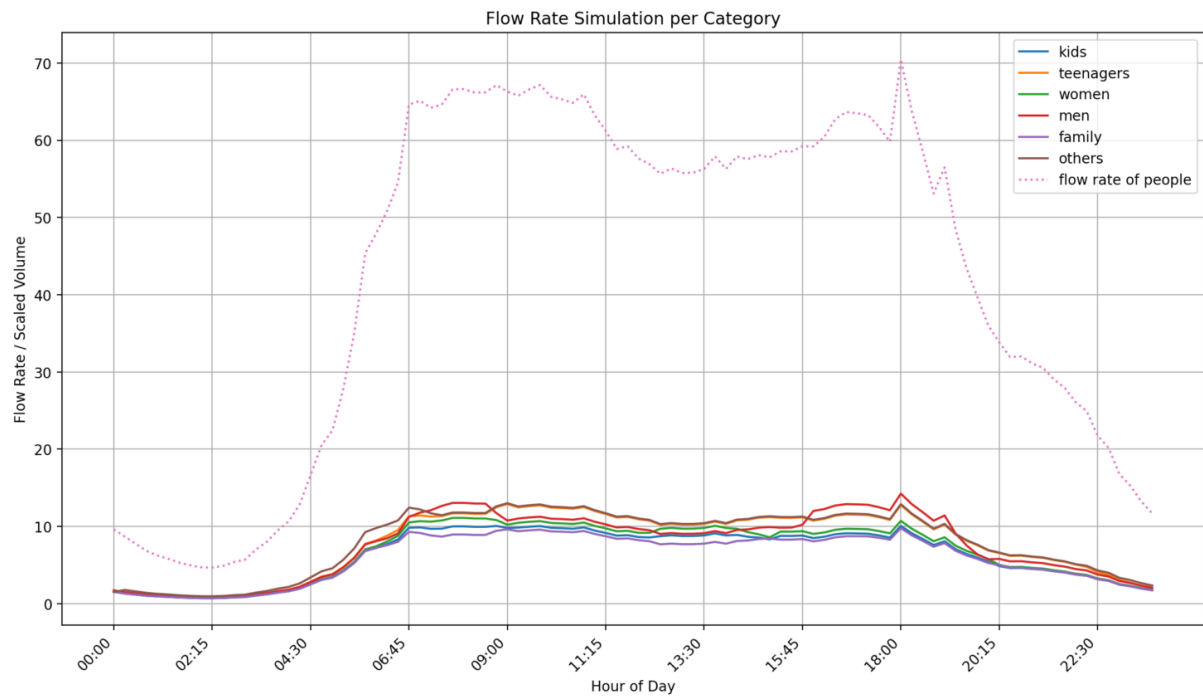


Fairness Consideration in Digital Twin

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Abstract

The goal of this project is to develop a digital twin of the socio-economic landscape of students and their families in the city of Bologna. This digital twin will serve as a simulation and analysis tool to understand how different socio-economic groups interact with the city's infrastructure—particularly at key access points such as city entrances, transportation hubs, and restricted traffic zones (ZTL). By estimating mobility-related statistics at these access points and linking them to socio-economic profiles, the system aims to simulate how changes in urban mobility policies—such as new ZTL restrictions, car access tariffs, or public transportation adjustments—affect the behavior and movement patterns of different population segments. This tool will help policymakers evaluate the social and economic impact of urban planning decisions, promoting more equitable and data-informed approaches to mobility management in Bologna.

1 Introduction

This project is a component of the broader “Digital Twin for Fairness in Bologna” initiative, which aims to create comprehensive digital representations of the city's physical and social infrastructure. Urban mobility policies directly impact citizens' daily lives, with different socio-economic groups experiencing varying effects. Our specific contribution develops a computational framework to analyze and visualize how Bologna's city infrastructure—specifically its access points, schools, and green areas—relates to population movement patterns.

What distinguishes this project is its explicit focus on fairness evaluation. The system allows policymakers to assess potential urban changes not only for efficiency but for their equitable impact across population segments. For example, a modification that might improve overall traffic flow but disproportionately penalizes families with children could be identified and then flagged as potentially unfair.

The project integrates multiple data sources including:

- City gates (Porte) locations and traffic flow data
- Educational institutions' geographical distribution
- Green areas and public spaces
- Traffic flow rates at city gates

By creating an interactive visualization system, the project enables policymakers to:

1. Identify mobility patterns at key city gates
2. Analyze relationships between city infrastructure and population segments
3. Simulate potential impacts of policy changes on different socio-economic groups
4. Support evidence-based decision making for equitable urban planning

2 Background and Related Work

Bologna, a historic city with origins dating back to the Etruscan civilization, presents unique urban mobility challenges stemming from its medieval urban layout. The city center, contained within ancient walls, features narrow streets and limited road capacity that were designed for pedestrians and horse-drawn vehicles rather than modern transportation.

The city has implemented several Restricted Traffic Zones (ZTL) which limit vehicle access based on time of day, vehicle emissions standards, and special permits. While effective for preserving historical areas and reducing congestion, these access restrictions create differential impacts across socioeconomic groups:

- Families with children often need more flexible transportation options to manage school commutes, extracurricular activities, and errands
- Lower-income residents who live in peripheral areas with limited public transportation may face disproportionate barriers to accessing central city services
- Small business owners and service providers requiring frequent access to the city center bear additional costs for special permits

Bologna also faces unique geographical constraints, situated in a valley surrounded by hills, which limits expansion possibilities and creates natural bottlenecks at city gates (porte). These access points become critical control points for traffic management but also potential points of inequity if not managed with fairness considerations.

2.1 Digital Twins for Urban Planning

A digital twin is a virtual replica of physical assets, processes, or systems that can be used for simulation, analysis, and optimization. In urban contexts, digital twins integrate multiple data sources to create comprehensive models that enable planning scenarios and real-time monitoring.

Early urban digital twins focused primarily on physical infrastructure and traffic flow optimization, such as:

- Singapore’s Virtual Singapore platform, which creates a data-rich 3D model of the entire city-state [3]
- Amsterdam’s 3D Digital Twin, focusing on spatial planning and development visualization [4]

More recent developments have expanded digital twin capabilities to include:

1. **Multi-layer integration:** Combining transportation networks, land use patterns, environmental factors, and demographic information
2. **Dynamic simulation:** Modeling time-based changes and system interactions rather than static representations
3. **Feedback mechanisms:** Incorporating real-time sensor data to continuously update and refine models
4. **Policy testing:** Evaluating the impacts of potential interventions before implementation

However, most urban digital twins still prioritize physical and operational efficiency over social equity considerations, creating a gap that this project aims to address.

2.2 Fairness in Urban Planning and Mobility

Mobility is particularly critical for fairness considerations because it acts as a gateway to other opportunities and services. In Bologna specifically, the implementation of ZTL policies has raised fairness concerns. While beneficial for environmental quality and historical preservation, these restrictions may create disproportionate burdens for:

- People who cannot afford to live in the city center but need to access services there
- Those with mobility impairments requiring door-to-door transportation
- Families with complex mobility patterns involving multiple stops (schools, workplaces, shopping)

2.3 Computational Approaches to Socioeconomic Analysis

Computational methods for analyzing socioeconomic factors in urban settings have evolved significantly in recent years. Relevant approaches include:

1. **Agent-based modeling:** Simulating individual decision-making behavior to understand emergent patterns
2. **Spatial accessibility analysis:** Quantifying access to services based on travel time, distance, and transportation modes
3. **Mobility pattern analysis:** Using GPS data, mobile phone records, or transportation surveys to identify movement patterns
4. **Socioeconomic segregation metrics:** Measuring how urban form and transportation infrastructure contribute to or mitigate segregation

These computational approaches enable more nuanced understanding of how urban systems affect different population segments.

3 Methodology

Our approach to developing the digital twin of Bologna’s socio-economic landscape integrated multiple analytical perspectives, weaving together geospatial analysis, demographic modeling, and fairness evaluation frameworks into a cohesive methodology. Rather than simply mapping physical infrastructure, we sought to understand how diverse population segments experience and navigate Bologna’s urban environment differently.

3.1 Conceptual Framework

The foundation of our methodology lies in recognizing that urban mobility is inherently tied to questions of fairness and equity. Our framework explicitly models these differences by considering how infrastructure characteristics affect distinct demographic segments:

Families with children navigate complex mobility patterns involving school commutes, shopping, and recreational activities, often requiring direct access to multiple services within constrained time windows. Young children depend primarily on safe pedestrian access to educational facilities and green spaces. Teenagers exhibit higher price sensitivity and greater reliance on public transportation. Adult men and women often display different mobility patterns, with women more frequently managing household errands and childcare-related travel alongside professional commitments. Adult men and women can belong to a family, but, when talking about the single category, it means that they are not moving as the whole family. The "others" category serves as a catch-all group representing individuals who do not clearly fit into the chosen demographic classifications, ensuring comprehensive coverage of the urban population.

To quantify these differential experiences, we developed a multi-dimensional weighting system capturing how various urban attributes affect each demographic group. For instance, proximity to public schools carries significant weight for children and families but minimal importance for young adults without children. This weighting approach allows for nuanced assessment of how urban infrastructure changes might benefit certain groups while disadvantaging others.

3.2 Data Integration and Spatial Representation

The digital twin integrates three key datasets to create a comprehensive representation of Bologna’s urban landscape. First, we mapped the city gates (*porte*) that function as critical control points

for urban mobility. Each gate was associated with specific geographical coordinates and infrastructure attributes reflecting factors such as entry cost, proximity to services, and transportation alternatives. The gates provide natural focal points for analyzing how mobility policy changes might affect different population segments.

Second, we incorporated educational facilities throughout Bologna, including 58 distinct institutions ranging from early childhood centers to secondary schools. The educational data distinguishes between public and private institutions, recognizing that these represent different accessibility considerations for various socio-economic groups. Third, we mapped green spaces and recreational areas, similarly distinguishing between freely accessible parks and fee-based facilities.

3.3 Accessibility Analysis

Defining accessibility required moving beyond simple distance metrics to consider how different demographic groups experience urban space. We developed a range-based analysis framework centered on city gates, with configurable radius typically set between 650-750 meters to represent approximately 10-minute walking distances.

This range-based approach reveals important patterns in service distribution across Bologna. For instance, our analysis showed that the Saragozza gate provides access to both educational institutions and green spaces within walking distance, while San Vitale has fewer nearby schools but better public transportation connections. These patterns directly influence the effort required from different demographic groups to access essential services—a key component of our fairness assessment.

3.4 Temporal Dimension

Urban mobility patterns exhibit strong temporal variations that significantly impact accessibility. Our methodology incorporates this temporal dimension through probabilistic models representing how different demographic groups utilize city infrastructure throughout the day.

For families, we modeled peak mobility during morning school hours (7:30-9:00) and evening return times (17:00-20:00), with significantly reduced midday presence. Children’s mobility concentrates primarily during school transport hours, while adult patterns distribute more broadly across workday hours.

The integration of temporal patterns into our accessibility model allows for more realistic simulation of how policy changes might affect different groups at different times. For instance, a restriction implemented during school hours would disproportionately impact families with children, while evening restrictions would have different demographic effects.

3.4.1 Traffic Flow Time Series Analysis

For each gate (porta), we aggregated time-stamped traffic data into a continuous time series representing how many people cross each access point throughout the day. Rather than distinguishing between vehicle types, we focused on overall traffic volume as a proxy for human mobility patterns. This approach allowed us to identify distinct temporal signatures for different city gates.

By averaging these patterns across multiple days and analyzing their consistency, we constructed representative daily time series for each gate. This data reveals important city-wide patterns:

- Gate-specific temporal signatures that reflect their functional role in the city’s mobility network
- Consistent peak times across the city, suggesting synchronized movement patterns

3.4.2 Demographic Mobility Models

To connect these empirical traffic patterns with population segments, we developed temporal probability functions representing how different demographic groups utilize city infrastructure throughout the day. These functions model the likelihood of different user categories being present in the overall traffic at specific times.

For families with children, our model captures the characteristic "double peak" pattern tied to school schedules:

$$P_{family}(t) = \begin{cases} 0.7 & \text{if } t \in [7.5, 8.5] \cup [16, 19] \\ \text{ramp_up}(0.3, 0.7) & \text{if } t \in [6, 7.5] \cup [15.5, 16] \\ \text{ramp_down}(0.7, 0.3) & \text{if } t \in [8.5, 9] \cup [19, 20] \\ 0.3 & \text{if } t \in [9, 15.5] \cup [20, 24] \\ 0.2 & \text{otherwise} \end{cases} \quad (1)$$

Similar functions model temporal patterns for other demographic groups, reflecting their distinct daily routines.

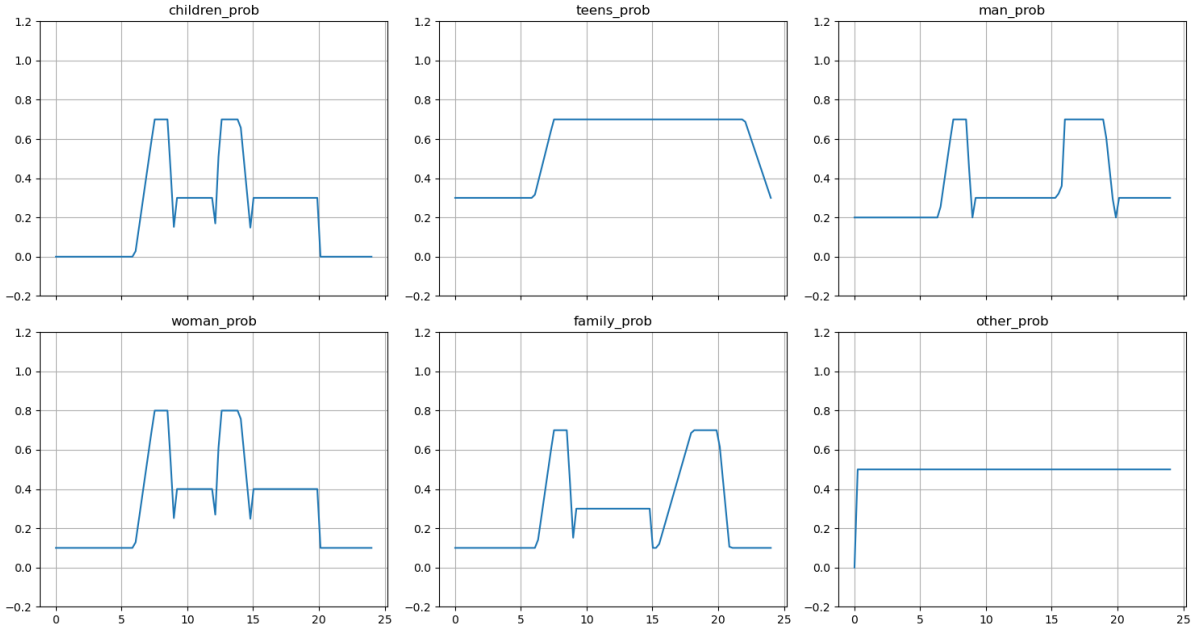


Figure 1: $P(category == outside|t)$ Probability that a certain category (e.g. children) is outside given the time.

The integrated analysis of aggregate gate flow data and demographic probability models allows us to estimate how different population segments experience accessibility across the day. This temporal dimension is critical for fairness assessment, as it reveals how policy changes at specific times might disproportionately affect certain groups. For example, a ZTL restriction implemented during school drop-off hours (8:00-8:45) would have maximum impact on families with children, while similar restrictions during midday would distribute impacts more evenly across demographic groups.

3.5 Fairness Evaluation Framework

At the heart of our methodology lies a framework for evaluating the fairness implications of potential urban mobility policies. Central to this framework is the concept of effort, which we

model as a cost function that quantifies, in a simplified yet meaningful way, the accessibility burden experienced by each demographic category. This function allows us to systematically compare and analyze the relative "cost" or difficulty of access for different groups under various policy scenarios.

The evaluation process begins by establishing baseline "effort scores" for each demographic group at each city gate. These baseline scores reflect the current accessibility landscape and serve as reference points for measuring the impact of proposed changes.

At the beginning the effort score calculation integrates spatial, attribute-based, and demographic factors. For each city gate, infrastructure attributes are weighted according to demographic-specific importance factors. When simulating policy changes, the system recalculates effort scores based on modified attribute values (according to 2).

$$Effort_{category}(\theta) = \sum_{feature \in features} door_settings(\theta)_{feature} * features(category)_{feature} \quad (2)$$

This approach permits evidence-based assessment of equity considerations alongside traditional efficiency metrics when evaluating urban mobility policies. Instead of simply optimizing for aggregate traffic flow or revenue generation, planners can identify and mitigate unfair burdens on specific demographic segments.

3.6 Visualization and Analytical Approach

To support both technical analysis and stakeholder communication, we developed an interactive visualization system that makes complex spatial relationships accessible to non-technical stakeholders. The multilayered map visualization allows exploration of relationships between city gates, schools, and green areas, while dynamic charts illustrate the demographic impacts of potential policy changes.

This visualization approach serves both analytical and communication functions. Analytically, it enables exploration of spatial patterns and identification of service gaps. For communication, it makes fairness considerations visually apparent alongside traditional infrastructure representations, supporting more inclusive discussion of urban planning alternatives.

4 Implementation

This section presents the mathematical foundations underlying our demographic probability distribution model. The model aims to capture the temporal variations in the presence of different demographic groups in public spaces throughout a 24-hour cycle, which is essential for a fair evaluation of these systems.

4.1 Elementary Functions

The core mathematical components of our model are two elementary functions that enable flexible representation of temporal patterns:

Constant Value Function:

$$f_{\text{const}}(t; v, t_0, t_1) = \begin{cases} v, & \text{if } t_0 < t \leq t_1 \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

Linear Ramp Function:

$$f_{\text{ramp}}(t; v, t_0, t_1) = \begin{cases} v \cdot \frac{t-t_0}{t_1-t_0}, & \text{if } t_0 < t \leq t_1 \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

These functions serve as building blocks for constructing complex temporal probability distributions.

4.2 Composite Probability Functions

For each demographic category c , we define a probability function $P_c(t)$ as a composite of multiple elementary functions:

$$P_c(t) = \text{const} + \sum_i f_{\text{const},i}(t) + \sum_i f_{\text{ramp},i}(t) \quad (5)$$

This formulation allows us to model:

- Gradual increases in probability (e.g., morning commute hours)
- Stable probability plateaus (e.g., school/work hours)
- Gradual decreases in probability (e.g., evening return home)
- Combinations of these patterns to capture the complex daily rhythms of urban demographics

NB. const is a small value > 0 which enables, for each category to not have a null probability (Laplace smoothing)

4.3 Model Parameterization

The model parameters were calibrated through assumptions we made about typical urban temporal patterns:

- **Children:** Higher probabilities during school commute hours (7:30-8:30, 12:30-14:00) with moderate presence during afternoon hours
- **Teenagers:** Relatively constant presence throughout day with slightly higher values during afternoon/evening
- **Men:** Bimodal distributions capturing morning and evening commutes, with pronounced peaks during typical workday start/end times
- **Women:** Multimodal distribution capturing complex daily patterns including:
 - Morning responsibilities (bringing children to school, 7:30-9:00)
 - Midday activities (household errands, shopping, part-time work, 12:00-14:00)
 - Afternoon peaks (picking up children from school, household responsibilities, 15:00-16:00)
- **Families:** Higher probabilities during morning commutes and extended higher probability during evening/leisure hours (15:00-20:00)
- **Others:** Uniform distribution modeling the behavior of multiple other categories of people, for this reason there is no other assumption to be made about this distribution

Each demographic category has independently parameterized functions reflecting their distinct temporal patterns, while maintaining demographic diversity throughout the 24-hour cycle.

4.4 Computational Framework for Demographic Flow Simulation

The simulation of demographic flows through urban gates requires a comprehensive computational process that integrates the probability models with actual gate crossing data. This process allows us to estimate the demographic composition of pedestrian flows at different times of day.

4.4.1 From Raw Counts to Demographic Distributions

Given the total count of people crossing a gate at time t , denoted as $N(t)$, we use our probabilistic model to estimate how many individuals from each demographic category are represented in this count. The process involves the following mathematical steps:

1. **Effort-to-Probability Conversion:** We first convert effort values $Effort_{category}$ for each category c into base probabilities:

$$P_{\text{base}}(c) = 1 - \frac{Effort_{category}}{\text{MAX_VAL}} \quad (6)$$

where MAX_VAL is a normalization constant (set to 5 in our implementation). This creates an inverse relationship between effort and probability—categories requiring more effort to access a location have lower probabilities of being present.

2. **Time-Dependent Scaling:** These base probabilities are then scaled by time-dependent factors derived from our probability functions:

$$P_{\text{scaled}}(c, t) = P_{\text{base}}(c) \cdot P_{\text{time}}(c, t) \quad (7)$$

where $P_{\text{time}}(c, t)$ represents the time-dependent probability function for category c at time t .

3. **Probability Normalization via Softmax:** To ensure that probabilities across all demographic categories sum to 1 at any given time, we apply the softmax function:

$$P_{\text{final}}(c, t) = \frac{e^{P_{\text{scaled}}(c, t)}}{\sum_{c' \in \text{Categories}} e^{P_{\text{scaled}}(c', t)}} \quad (8)$$

4. **Computing Demographic Counts:** Finally, we multiply the normalized probabilities by the total observed count to estimate the number of individuals from each category:

$$N(c, t) = N(t) \cdot P_{\text{final}}(c, t) \quad (9)$$

This computational process allows us to transform aggregate pedestrian counts into demographically disaggregated estimates based on our time-dependent probability models and effort parameters.

4.4.2 Relationship Between Effort and Probability

The mapping between effort values $Effort_{category}$ and base probabilities $P_{\text{base}}(c)$ deserves special attention. The effort parameter represents the relative difficulty or barrier to entry for a particular demographic category at a given location. Higher effort values indicate greater barriers, which naturally correspond to lower probabilities of presence.

5 Results and Analysis

Our dashboard reveals how Bologna’s urban characteristics shape demographic patterns through the interplay of environmental parameters and demographic weights. This approach captures nuances that static analyses miss entirely.

Adjusting the "aree verdi gratuite" parameter reveals the significance of Bologna’s parks like Giardini Margherita across multiple demographic groups. While families show the strongest response (-1.7 weight), we observed that young adults also demonstrate a substantial sensitivity (-1.2 weight), reflecting Bologna’s vibrant student population of over 80,000 who use these spaces as important social gathering points. When green space accessibility improves in our simulation, both families and young adults respond positively, though with different magnitudes—a pattern that aligns with Bologna’s observed social dynamics where parks serve distinct but equally important functions for different categories.

When manipulating the "scuole pubbliche" parameter, we observe Mascarella and San Vitale showing amplified responses that reflect the lower concentration of educational institutions in these areas. The data reveals a gendered pattern in school proximity sensitivity—women’s presence increases more significantly than men’s when school accessibility improves (weights -1.3 versus -0.2), reflecting differences in childcare responsibilities.

Economic parameters demonstrate how entrance costs affect demographic flows differently. Families show higher sensitivity to cost changes (weight 0.9), while teenagers actually display a negative weight (-0.8), suggesting they might be more likely to congregate in areas with controlled access—possibly reflecting their preference for structured social spaces over completely open ones.

The mobility parameter in our model captures Bologna’s distinctive urban layout with its extensive porticoed walkways and limited traffic zones. The values in our dashboard (-0.9 across all gates) provide a baseline that interacts with demographic-specific sensitivities to mobility: women (-1.4), men (-1.3), and children/families (both -1.2). These differences, though subtle, generate variations in demographic flows when mobility conditions change.

These findings show how people move around Bologna based on lots of different factors that all work together - like where parks and schools are located, what time of day it is, and who people are (kids, students, adults, families).

By giving different weights to these factors for each demographic group, our digital twin can simulate how Bologna’s unique features - its historic center, university areas, and walkable streets - actually affect real people’s daily movements. This is super helpful for city planners and policymakers who want to create more fair and inclusive cities. The digital twin helps them understand the complex patterns of how different groups navigate urban spaces, which means they can make better decisions that work for everyone who lives in Bologna.

6 Discussion

Our findings suggest several important considerations for urban mobility policy:

1. **Equity in Access:** The spatial distribution of city gates relative to schools reveals potential disparities in access to educational facilities.
2. **Traffic Management:** Flow rate patterns indicate opportunities for targeted traffic management during peak hours.
3. **Green Space Accessibility:** The analysis identifies areas where green space accessibility could be improved, particularly in relation to educational facilities.
4. **Fairness-Centered Planning:** The ability to simulate impacts across different demographic groups enables the rejection of policies that might create unfair burdens on specific segments of the population.

6.1 Limitations

Our project has some important limitations that we need to acknowledge. First off, the weights we assigned to different demographic groups are mostly based on our best guesses rather than actual research specific to Bologna. While we think our assumptions make sense, we really should verify them with surveys and real mobility studies in the future.

Another issue is that we just used straight-line distances in our accessibility calculations instead of actual street routes. This probably makes places seem more accessible than they really are, especially in areas with weird street layouts or where there are physical barriers like tram tracks or construction sites. A better version would use actual walking paths and consider stuff like whether sidewalks exist and if areas are safe to walk through.

We also simplified people into just a few categories (kids, students, men, women, families, others), which doesn't capture all the differences within these groups. People with different incomes, disabilities, or family situations might have totally different needs and movement patterns. The "others" category is used to represent all individuals who do not fall into the selected categories. Even though this is a simplification, it provides an easy way to account for both frequent and rare cases, which can be difficult to assign to a specific category.

The probabilities in our model should ideally be estimated using real traffic data from the city rather than our mathematical functions. The number of people crossing each gate should come from a proper simulator instead of just averaging historical data, which would better account for daily and seasonal variations. A bigger advantage of using a simulator would be the ability to take into account each door parameters to better simulate the number of people crossing the door.

Some of these parameters could potentially be estimated through machine learning, like using a neural network trained on actual movement data to predict how different demographic groups respond to environmental factors. This would give us a more data-driven approach instead of relying so much on our assumptions.

Despite these limitations, our model still demonstrates the key concepts and provides a starting point for more sophisticated fairness analysis in urban environments.

6.2 Future Work

While our current implementation provides valuable insights, several limitations and opportunities for enhancement exist:

1. **Categories:** New and different categories can be used to improve the analysis or to model specific segments of the population. These categories can also be chosen based on the datasets that will be available in the future.
2. **Estimating probability functions:** The temporal probability functions introduced in 3.4.2 could be estimated by training Gaussian Mixture Models (GMMs) on real-world traffic data annotated by user categories. This data-driven approach would allow us to capture more nuanced and empirically grounded temporal patterns of infrastructure usage across demographic groups.
3. **Sociological Validation:** The current model makes assumptions about user preferences and behaviors without comprehensive sociological research. Future work should validate these assumptions through surveys, interviews, and observational studies.
4. **Socio-economic Data Integration:** Future iterations should incorporate more granular socio-economic data to better represent population segments.
5. **Statical areas:** Replace the current gate-based approach with statistical areas as the fundamental spatial units. This would enable a more comprehensive and demographically

coherent analysis of mobility patterns across the city. Importantly, our modular codebase allows for a straightforward implementation of this change, facilitating the integration of statistical areas without major structural modifications.

6. **Temporal Analysis:** Extending the model to better capture time-based variation in movement patterns.
7. **Transportation Network Integration:** Incorporating public transportation routes and schedules would enhance the mobility analysis.
8. **Policy Simulation:** Developing more sophisticated simulation capabilities to model the impact of specific policy changes.
9. **Complex Simulation:** The simulation can be composed of multiple models, each responsible for simulating a specific parameter for a category, or the overall flow rate of people at a city gate depending on the day of the year. This modular approach allows specific parts of the simulation to be modeled independently, while using the main framework as a controller to coordinate and implement various complex functionalities in a unified environment.

7 Conclusion

For our project, we created a digital twin framework that shows how people move around Bologna and how fair the city is for different groups of people. As part of the broader “Digital Twin for Fairness in Bologna” initiative, it specifically contributes tools for evaluating the equitable impact of urban planning decisions. It can be easily integrated with past work to create a more sophisticated simulator.

The integration of spatial data on city gates, educational institutions, and green areas provides a foundation for understanding how urban infrastructure and access policies impact different population segments. By focusing explicitly on fairness considerations, the system allows planners to identify and avoid changes that would disproportionately disadvantage vulnerable groups like families with children.

The interactive visualization tools we developed let stakeholders see how demographic distributions change in response to environmental adjustments, which supports evidence-based decision-making for urban development projects.

Going forward, this framework can be extended to incorporate additional data sources, more sophisticated simulation capabilities, and rigorous sociological validation, further enhancing its utility for evidence-based policy development in Bologna and potentially other urban environments.

8 References

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